Adaptation Experiment 1: Univariate Control

Thesis experiment note

1 Theoretical overview and goal

A forecasting adapter is useful only if it improves a well-defined, untouched backbone. For a look-back $x \in \mathbb{R}^L$, a pretrained forecaster produces $\hat{y}_0 = f(x) \in \mathbb{R}^H$. The univariate control evaluates f directly, without retrieved neighbors, gates, or trainable adaptation. It therefore separates backbone error from error introduced by the retrieval and adaptation pipeline.

The main metrics are

$$\mathrm{MSE} = H^{-1}\|\hat{y}_0 - y\|_2^2, \qquad \mathrm{nMSE} = H^{-1}\left\|\frac{\hat{y}_0 - y}{\sigma(x)+\varepsilon}\right\|_2^2.$$

Per-user aggregation and the worst 10% user mean reveal whether a good global mean hides systematic failures on a subset of series.

Experimental goal. Establish the no-adaptation reference, measure its temporal and user-wise heterogeneity, and provide the denominator against which every retrieval method must be compared.

2 Implementation details

Dates are split chronologically as $(T_0, T_1, T_2, T_3) = (0.30, 0.35, 0.15, 0.20)$. The control command maps T_1, T_2, and T_3 to the labels `train`, `oracle`, and `eval`; the Slurm job evaluates the held-out T_3 period. Each query contains all users at one date, but the backbone is called as a univariate model with one channel per sample.

Component	Current sweep value
Datasets	Electricity, Solar
Look-back–horizon	168–24, 672–168
Backbone	Chronos
Input normalization	Instance normalization
Evaluation stride	128 dates
Seed	1

Implementation: `src/experiments/experiment_univariate.py`
Launcher: `src/slurm/univariate.slurm`

It writes window-level losses, a JSON summary, a tensor payload, error distributions, horizon-error curves, user scatter plots, and prediction examples. The primary comparison is the same T_3 MSE/nMSE later used by adapted methods.

3 Interpretation

This experiment is descriptive rather than trainable. Any adapted method must beat the control on the same queries and aggregation rule. Improvements on T_1 or T_2 alone are insufficient evidence because those periods are used to fit adapters or gates.